CHAPTER

WAVE OPTICS

Topic-1

Interference

Topic-1

Interference

The light is a form of electromagnetic wave (radiation).

But, how do we know?

Basic features of the light wave :

- Interference
- Diffraction
- Polarization

CONCEPT

Topic-1 **Interference**

CONCEPT

Interference of Light

When two waves of exactly same frequency (coming from two coherent sources) travels in a medium, in the same direction simultaneously then due to their superposition, at some points intensity of light is maximum while at some other points intensity is minimum.

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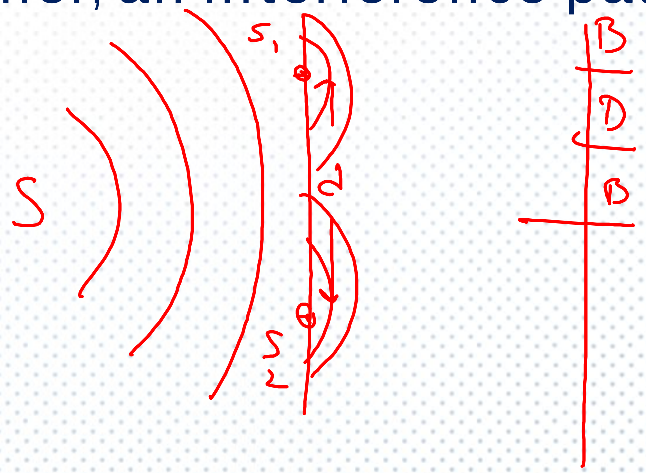
Interference

Young's Double Slit Experiment (YDSE)

Monochromatic light (single wavelength) falls on two narrow slits S_1 and S_2 which are very close together acts as two coherent sources,

when waves coming from two coherent sources superimposes on each other, an interference pattern is obtained on the screen.



DERIVATION

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Derivation :

Consider two waves having same frequency but slight different in phase and having same amplitude be y_1 and y_2

$y_1 = a\sin(\omega t)$; $y_2 = a\sin(\omega t + \phi)$

Let $\omega \rightarrow$ Frequency; $\phi \rightarrow$ Initial phase difference

Then according to principal of superposition.

$y = y_1 + y_2 = a\sin(\omega t) + a\sin(\omega t + \phi)$

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$= a[\sin(\omega t) + \sin(\omega t)\cos\phi + \cos(\omega t)\sin\phi]$

$= a\sin(\omega t)\,(1 + \cos\phi) + a\cos(\omega t)\,(\sin\phi)$

Replacing,

$a(1 + \cos\phi) = R\cos\theta$ ...(i)

$a\sin\phi = R\sin\theta$ ...(ii)

Then $y = R\sin\omega t\cos\theta + R\cos\omega t\sin\theta$

$y = R\sin(\omega t + \theta)$

Interference

The final equation represents the interference pattern on the screen.

Squaring equation (1) and (2)

$R \cos\theta = a(1 + \cos\phi)$

$R \sin\theta = a \sin\phi$

$$R^2 \sin^2\theta + R^2 \cos^2\theta$$

$$= a^2[(1 + \cos\phi)^2 + \sin^2\phi]$$

$$R^2 = a^2(1 + \cos^2\phi + 2\cos\phi + \sin^2\phi)$$

$$= \boxed{R^2 = 4a^2 \cos^2(\phi/2)}$$

Intensity is proportional to square of amplitude

$$\boxed{I = R^2 = 4a^2 \cos^2(\phi/2)}$$

Condition for bright band :

If $\cos^2(\phi/2) = 1$ then

$I = 4a^2 \Rightarrow \Delta\phi = 0, 2\pi, 4\pi \ldots$ Phase difference = $2n\pi$

$$\boxed{\Delta\phi = \frac{2\pi}{\lambda}\Delta x}$$

$$I = 4a^2\cos^2(\phi/2)$$

$$2n\pi = \frac{2\pi}{\lambda}\Delta x$$

$\boxed{\Delta x = n\lambda}$ (Path difference)

$n = 0, 1, 2 \ldots$

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Condition for dark region :

$I = 0 \Rightarrow \cos^2(\phi/2) = 0$

$\Rightarrow \Delta\phi = \pi, 3\pi, 5\pi \ldots$ Phase difference $= (2n - 1)\pi$

$\Delta\phi = \frac{2\pi}{\lambda}\Delta x$

$\underline{I = 0}$

$\Rightarrow (2n - 1)\pi = \frac{2\pi}{\lambda}\Delta x$

$\Rightarrow \Delta x = \frac{(2n-1)\lambda}{2}$

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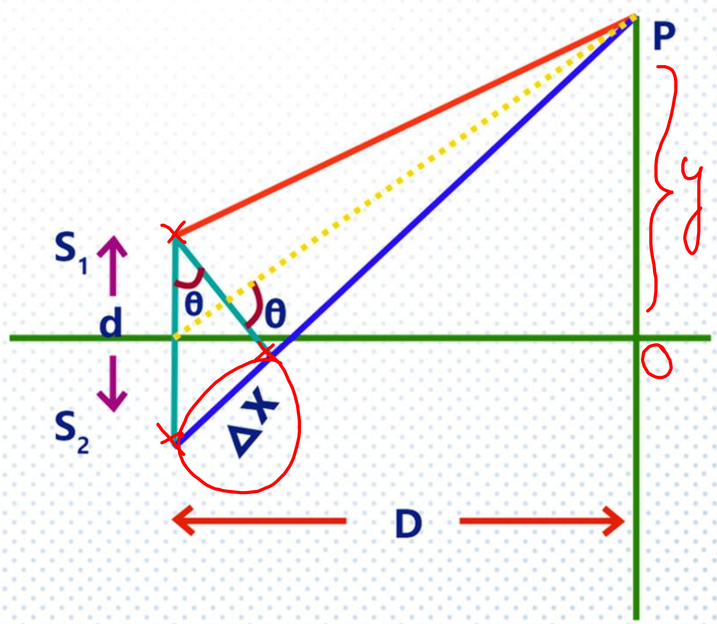
Let ‘d’ is the distance between slits and ‘D’ is the distance between slits and screen. Then the path difference by the two light wave is $\Delta x = d \sin\theta$

$$\sin\theta = \frac{\Delta x}{d}$$

$$\Delta x = d \sin\theta$$

$$y = y_1 + y_2 + \dots$$



EXAMPLES

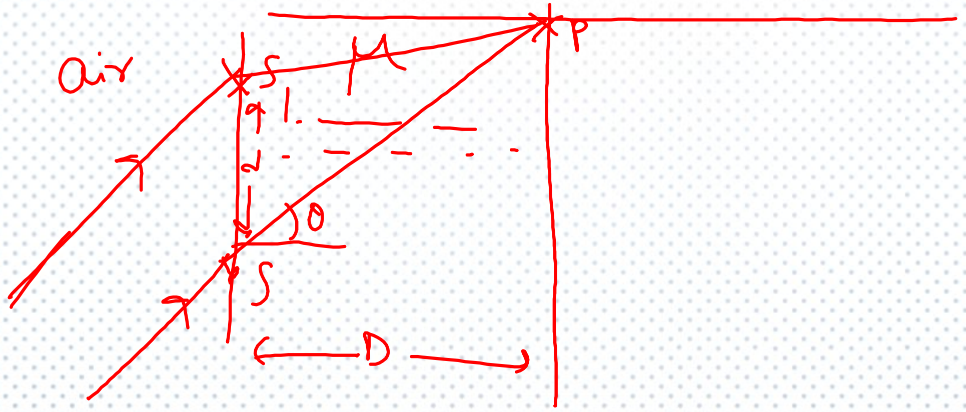
Topic-1

Interference

Q1

Light is incident at an angle ϕ with the normal to a vertical plane containing 2 narrow slits S_1 and S_2 at a separation d.

The medium to the left of the slit plane in air and wavelength of incident light is λ. The medium to the right of the slit plane has refractive index μ. Find all values of angular position (θ) of a point 'P' where we will observe constructive Interference



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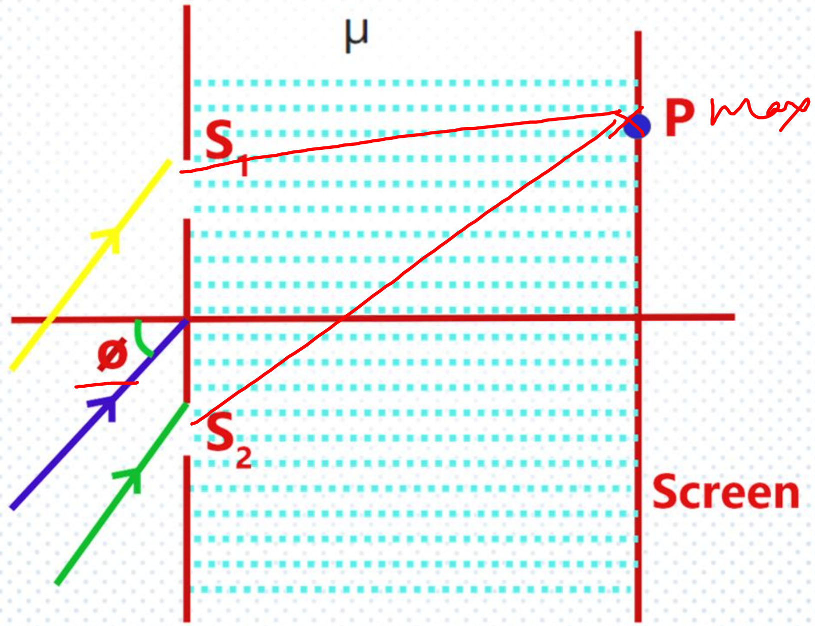
Interference



Sol :

Path difference between light incident on two slits is

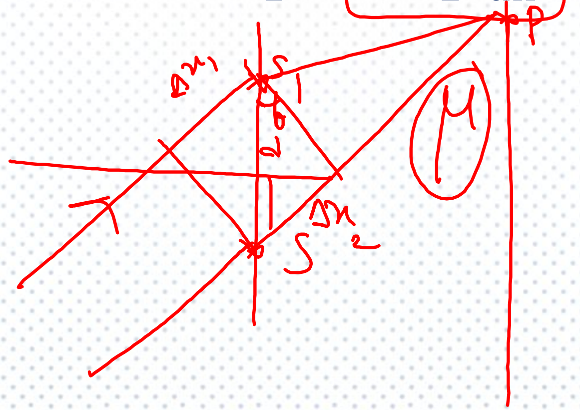
$\Delta x_1 = d \sin\phi$

Geometric path difference between S_2P and S_1P is $\Delta x_2 = d\sin\theta$

Equivalent in air is $(\Delta x_2)_{air} = \mu d \sin\theta$

Therefore total path difference is $\Delta x = \Delta x_1 + \boxed{(\Delta x_2)_{air}}$

$= d\sin\phi + \mu d\sin\theta$



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For maximum; $\boxed{\Delta x = n\lambda}$; n=0, 1, 2, 3

$$d\sin\phi + \mu d\sin\theta = n\lambda$$

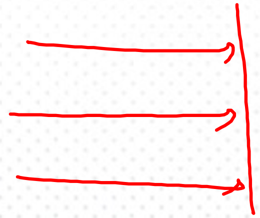
$$\mu d\sin\theta = n\lambda - d\sin\phi$$

$$\theta = \sin^{-1}\left[\frac{1}{\mu d}(n\lambda - d\sin\phi)\right]$$

Q2

Three narrow slits A, B and C are illuminated by a parallel beam of light of wavelength $'\lambda'$, P is a point on the screen exactly in front of point ‘A’. Slit plane is at a distance D from the screen $(D \gg \lambda)$. It is known that

$BP - AP = \frac{\lambda}{3}$



a) Find d in terms of D and λ

b) Write the phase difference between waves reaching at P from C and A

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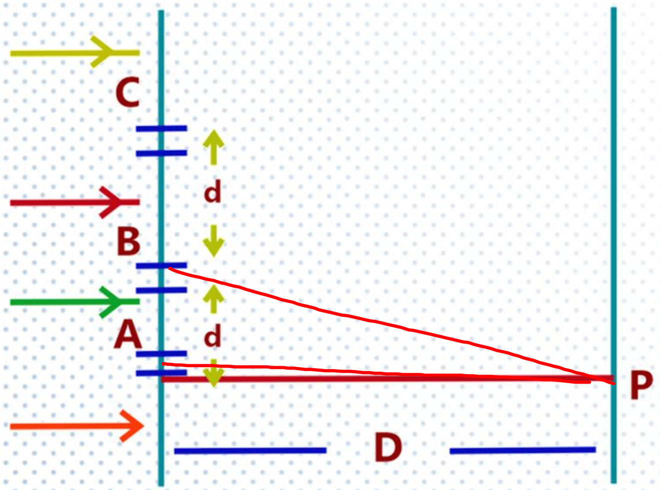
c) If intensity of 'P' due to any of the three slits individually is I_0. Find the resultant intensity at 'P'

$BP - AP = \Delta x$



Sol : (a) As given $BP - AP = \frac{\lambda}{3}$

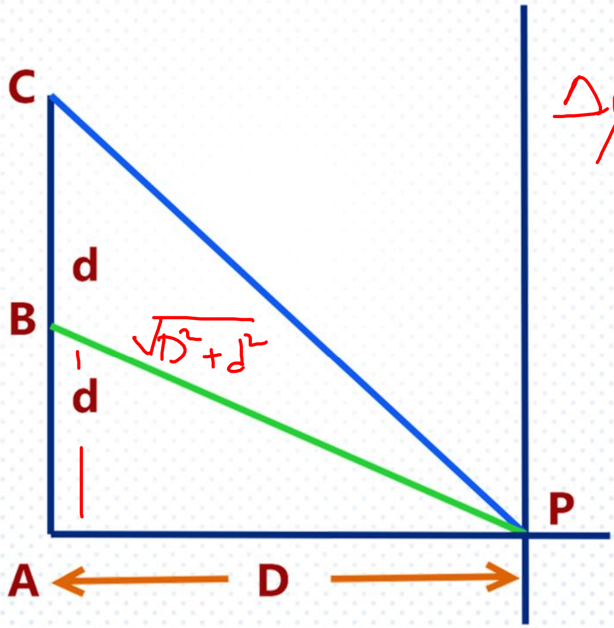
$$\sqrt{D^2 + d^2} - D = \frac{\lambda}{3}$$

$$= D\left[1 + \frac{d^2}{D^2}\right]^{1/2} - D = \frac{\lambda}{3}$$

$$= D\left[1 + \frac{d^2}{2D^2}\right] - D = \frac{\lambda}{3}$$

$d \ll D$ as $BP - AP = \frac{\lambda}{3} \ll D$

$$D + \frac{d^2}{2D^2} - D = \frac{\lambda}{3} \Rightarrow d = \sqrt{\frac{2D\lambda}{3}}$$



$$\Delta\phi = \frac{2\pi}{\lambda}\Delta x$$
$$= \frac{2\pi}{\lambda}\cdot\frac{\lambda}{3}$$
$$= \frac{2\pi}{3}$$

(b) Path difference for two waves

$$\Delta x_{CA} = CP - AP$$

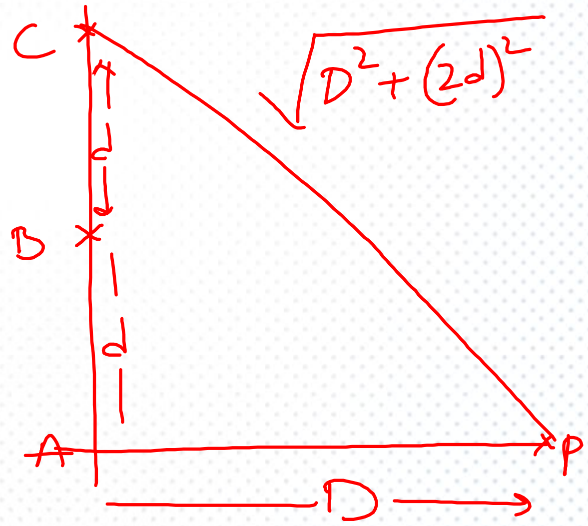
$$= \sqrt{D^2 + (2d)^2} - D$$

$$= D\left(1 + \frac{4d^2}{D^2}\right)^{1/2} - D = D\left(1 + \frac{2d^2}{D^2}\right) - D$$

$$= \frac{2d^2}{D^2}$$

$$= \frac{2}{D^2} \times \frac{2D\lambda}{3} = \frac{4\lambda}{3}$$

Phase difference $\Delta\phi_{AC} = \frac{2\pi}{\lambda} \times \frac{4\lambda}{3} = \frac{8\pi}{3}$



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(c) Path difference at P for waves A and B

$\Delta x_{AB} = \frac{\lambda}{3}$

$y = y_1 + y_2 + y_3$

$\Delta\phi_{AB} = \frac{2\pi}{\lambda} \times \frac{\lambda}{3} = \frac{2\pi}{3}$ and $\Delta\phi_{AC} = \frac{2\pi}{\lambda} \times \frac{4\lambda}{3} = \frac{8\pi}{3}$

So the wave equations are

$y_1 = a\sin(\omega t)$

$y_2 = a\sin\left(\omega t + \frac{2\pi}{3}\right)$

$y_3 = a\sin\left(\omega t + \frac{8\pi}{3}\right) = a\sin\left(\omega t + \frac{2\pi}{3}\right)$

Interference

Rizee THE PERFECT GUIDE

The resultant wave is $y = y_1 + y_2 + y_3$

$$= a\sin(\omega t) + 2a\sin\left(\omega t + \frac{2\pi}{3}\right)$$

$$A = \sqrt{a^2 + 4a^2 + 2.a.2a.\cos\left(\frac{2\pi}{3}\right)}$$

$$= \sqrt{3}\,a$$

$I \propto A^2$

So, Intensity $I = 3I_0$

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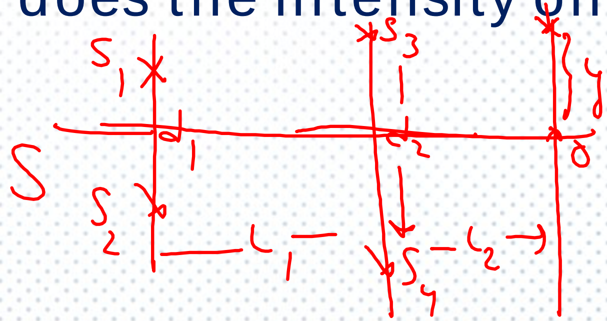
Q3

In the arrangement shown S in a point source of monochromatic light S_1 and S_2 are two slits located symmetrically with respect to the source with separation between then is d_1 parallel to this slit plane there are two more slits S_3 and S_4 at a separation d_2. There slits are also symmetrically located with respect to 'S'. A screen is located at a distance L_2 from this slit plane. How does the intensity on the screen with y and d_1 ?



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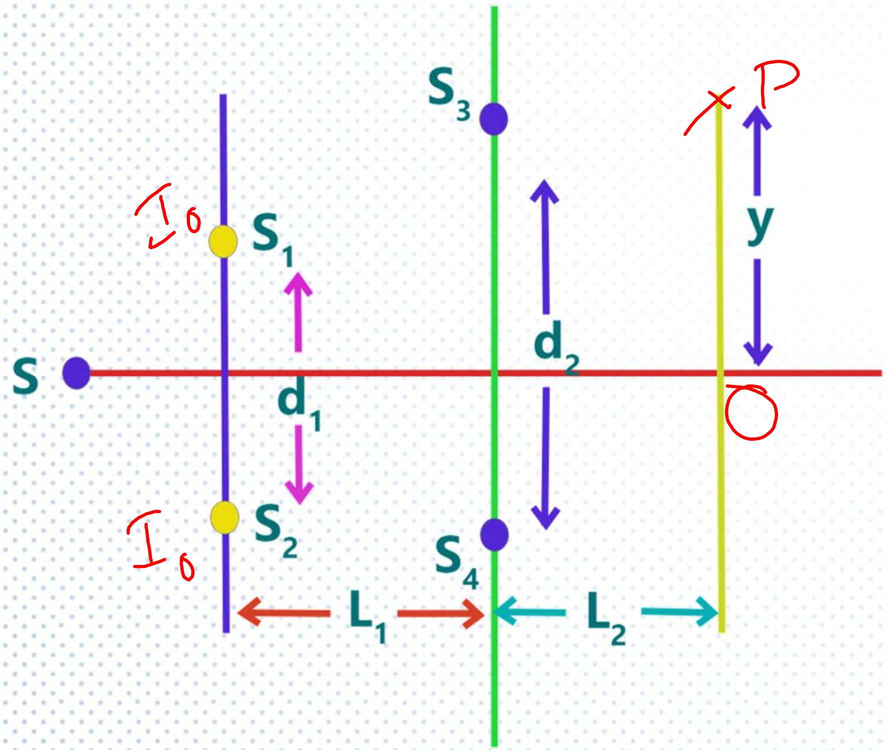

Sol :

Let I_0 is the intensity at S_3 and S_4 due to S_1 and S_2

Δx = to reach S_3 and S_4 = $d_1\left(\frac{d_2}{2L_1}\right)$

$$\Rightarrow \Delta\phi = \frac{2\pi}{\lambda}\frac{d_1 d_2}{2L_1} = \frac{\pi d_1 d_2}{\lambda L_1}$$

Then Intensity at S_3 and S_4

$$I = 4I_0 \cos^2\left(\frac{\pi d_1 d_2}{2\lambda L_1}\right)$$

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For the source S_3 and S_4

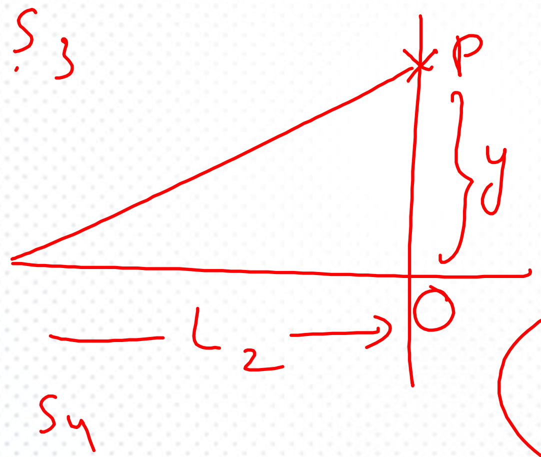
$$y = \frac{\Delta x' L_2}{d_2}$$

$$\Delta x' = \frac{d_2 y}{L_2}$$

$$\Delta\phi' = \frac{2\pi}{\lambda}\frac{d_2 y}{L_2} = \frac{2\pi d_2 y}{\lambda L_2}$$

Then $I' \propto I \cos^2\left(\frac{2\pi d_2 y}{\lambda L_2}\right)$

$$I' \propto 4I_0 \cos^2\left(\frac{\pi d_1 d_2}{2\lambda L_1}\right) \cos^2\left(\frac{2\pi d_2 y}{\lambda L_2}\right)$$



$\sin\theta = \frac{\Delta x}{d_2}$

$\tan\theta = \frac{y}{L_2}$

$y = \frac{\Delta x L_2}{d_2}$